Jimyung **Hong**

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EDUCATION

Mar 2023 – Feb 2027 (exp.)

Yonsei University — Seoul, Korea

B.E., Department of Artificial Intelligence

GPA 4.26/4.30 · **Major GPA 4.28/4.30** · **Ranked 1st in the Department**

Mar 2021 – Feb 2023

Korea University — Seoul, Korea

Division of Life Sciences (transferred; no degree)

PUBLICATIONS

Conference 2026

Training-free LLM Verification via Recycling Few-shot Examples

Dongseok Lee, **Jimyung Hong**, Dongyoung Kim, Jaehyung Kim

EMNLP 2026, Main Conference. Earlier version: Spotlight, ES-FoMo III Workshop at ICML 2025.

Preprint 2026

Diet Your LLM: Dimension-wise Global Pruning of LLMs via Merging Task-specific Importance Score

Jimyung Hong, Jaehyung Kim

arXiv:2603.23985

RESEARCH INTERESTS & EXPERIENCE

Interests

Efficient and adaptive AI systems; understanding and exploiting computational and representational redundancy through structured pruning, adaptive inference and model routing, and representation reuse across models.

Mar 2025 – Present

ML3 Lab, Yonsei University — Seoul, Korea

Undergraduate Researcher · Advisor: Prof. Jaehyung Kim

Research on efficient and reliable large language models; led a first-author project from problem formulation to manuscript and contributed to a co-authored EMNLP paper.

DIET: task-aware structured pruning of LLMs — First author

- Identified a gap between task-agnostic pruning and router-based task-aware methods, and formulated a training-free alternative.
- Built the full pipeline (activation profiling, importance voting, hard pruning) and ran large-scale evaluations across model families on a shared GPU cluster.
- Achieved **+2.1 pp** over SliceGPT at 20% sparsity on Gemma-2 2B, with **13.8%** lower peak memory in the hard-pruned model.
- Analyzed failure cases on other model families, which now motivate my follow-up question on when training-free pruning can be trusted; drafted the manuscript.

Training-free LLM verification — Second author · EMNLP 2026

- Implemented the method and its mathematical formulation, and designed the evaluation metrics.

Aug 2026 – Present

Cyber-Physical Intelligence Lab, McGill University — Montréal, Canada

Research Collaborator (remote) · Advisor: Prof. Xue (Steve) Liu

Remote collaboration on LLM serving systems; working with a graduate student on research questions I proposed.

Adaptive model routing for LLM serving — vLLM Semantic Router

- Studied which request signals are cheap enough to guide routing, and proposed conditioning the router on the deployment objective (quality, latency, cost).

Cross-model KV-cache transfer Ongoing

- Proposed research directions after reviewing the literature, then designed and ran controlled pilot experiments on an existing mapping toolkit (kvbridge).
- Now extending the pilots into an analysis of when, how, and why transfer works across source–target model pairs.

Sep 2026 – Present

Biomedical Ultrasound Lab, USC & MBIS Lab, DGIST

Los Angeles & Daegu

Prof. Qifa Zhou (USC) · Prof. Jae Youn Hwang (DGIST)

Early-stage project on wearable cardiac ultrasound, carried out with MBIS Lab and in ongoing discussion with Prof. Zhou's group.

Wearable ultrasound placement guidance Research scoping

- Formulated patch placement as a machine learning problem: localization from partial, noisy ultrasound observations.
- Scoped input representations (raw RF vs. B-mode) and data challenges: position labels, sparse supervision, and variation across patients and devices.

SELECTED HONORS AND AWARDS

2026 **Injaerim Program**, Korea Foundation for Advanced Studies (KFAS)

2025 **Presidential Science Scholarship**, Republic of Korea national merit; about 60 scholars per year

2025 Shin Dong-wook Academic Excellence Scholarship, Yonsei University

2024 – 2025 Highest Honor, Yonsei University College of Computing top 1%; three semesters

2024 Finalist, KOREN Net Challenge Season 11 top 10 teams

2023 3rd Place, Yonsei–Nexon RC Creative Platform of 100 teams

OTHER ACTIVITIES

Aug 2025 Two weeks

USC Viterbi School of Engineering

Los Angeles, CA

Completed two non-credit classes in AI and Data Science in association with the Integrated Media Systems Center.

SKILLS AND OTHER

ML & research Python, PyTorch, Hugging Face Transformers, lm-evaluation-harness; LLM evaluation and activation-profiling pipelines

Programming C/C++, SQL, Flask, Git, Linux

Languages Korean (native); English (**TOEFL iBT 110/120**, 5.5/6.0, Oct 2026)

Service Completed mandatory military service, Republic of Korea